

MD NURUZZAMAN SOJIB

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Education

PhD in Computer Science

August 2024 – Present

The University of Texas at El Paso

M.S. in Computer Science, expected Spring 2026

Bachelor of Science in Information Technology

March 2017 – February 2022

Jahangirnagar University

[Undergraduate Thesis](#)

Research Projects

Confidence-Aware Retrieval-Augmented Generation (RAG) for Cybersecurity

Ongoing

- Designing and evaluating RAG pipelines for cybersecurity Q&A with ChatGPT answers as baselines, where semantic similarity analysis showed about 12% lower performance than non-RAG, highlighting retrieval-answer misalignment.
- Advancing to entropy-based uncertainty modeling, with early results showing around 9% higher similarity than non-RAG, driving work on better confidence calibration and retrieval.

Predictive Modeling and Constrained Optimization for sustainable HFPVHP

Ongoing

- Developing an end-to-end forecasting pipeline with constrained optimization to support sustainable Hybrid Floating Photovoltaic-Hydropower Systems (HFPVHP) deployment on water surfaces.
- Implementing dual LSTM models for reservoir storage and outflow forecasting, with more than 35% training loss reduction on historical reservoir data.

Publications

- J. Alam, I. Hossain, M.T.A. Hossain, M.N. Sojib, O. Kosheleva, and V. Kreinovich, “**How to compare situations in which we measure different quantities with different uncertainty,**” *Proceedings of [SCAN2025](#)* (Accepted).

Teaching & Outreach Experience

Graduate Teaching Assistant

September 2024 – Present

The University of Texas at El Paso

- Supported courses: Advanced Object-Oriented Programming, Programming Language Concepts, and Computer Programming. (70 - 150 students per course)

Research Associate

Summer 2025

[CREEDS](#) (Sponsored by NSF)

- Designed and delivered interactive [workshops on Python and Machine Learning](#) for 12 high school teachers.
- Collaborated with faculty and graduate students to integrate cybersecurity and AI into STEM curricula.

Technical Skills

Languages: Python, C#, SQL

Machine Learning & Data Science Tools: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, MATLAB

Frameworks & Tools : ASP.NET Core, REST APIs, Docker

Academic Projects

Testing Prompt Engineering Techniques on LLM Based Phishing Detection

- Explored adversarial prompt engineering on LLM-based phishing detection, testing three attack strategies and four defense mechanisms on the Enron spam dataset using Google Gemini 2.0

Earthquake Prediction with Hill Climbing Hyperparameter Tuning

- Deep learning model for earthquake prediction, optimized using hill-climbing search over learning rate, batch size, and network depth.

Work Experience

Associate Software Engineer

March 2023 – June 2024

hSenid Business Solutions

- Developed automated HR configuration systems with ASP.NET Core and REST APIs, reducing setup time by 25%

Relevant Courseworks

- CS6350 - Advanced Algorithms
- CS6314 - Artificial Intelligence
- CS6386 - Software Architecture & Design
- CS5365 - Deep Learning
- CS6361 - Machine Learning

Professional Development & Academic Programs

ARITH Summer School 2025

May 1-2, 2025

32nd IEEE International Symposium on Computer Arithmetic

SLOAN Centers for Systemic Change Fellowship

2024–Present

The University of Texas at El Paso

Leadership & Achievements

- **General Secretary, Bangladeshi Student Association - The University of Texas at El Paso (2025–Present)**
- **Vice Chair, IEEE Jahangirnagar University Student Branch (2022–2023):** Organized technical workshops and hackathons.
- **Compensatory Scholarship Recipient (2018–2023):** Awarded by Jahangirnagar University for academic performance.
- Participated in PeoplesHR Hackathon 2023.